# HeatAtlas: Forecast Heat. Prioritise Relief.

Autonomous Extreme Heat Relief Engine, Constrained VRP Dispatch & Spatial Triage for Workers & NGOs

OFFICIAL HACKATHON TEAM

## Team Strawhats

TEAM CODE: 627JUH

**CHOSEN TRACK — THEME 2**

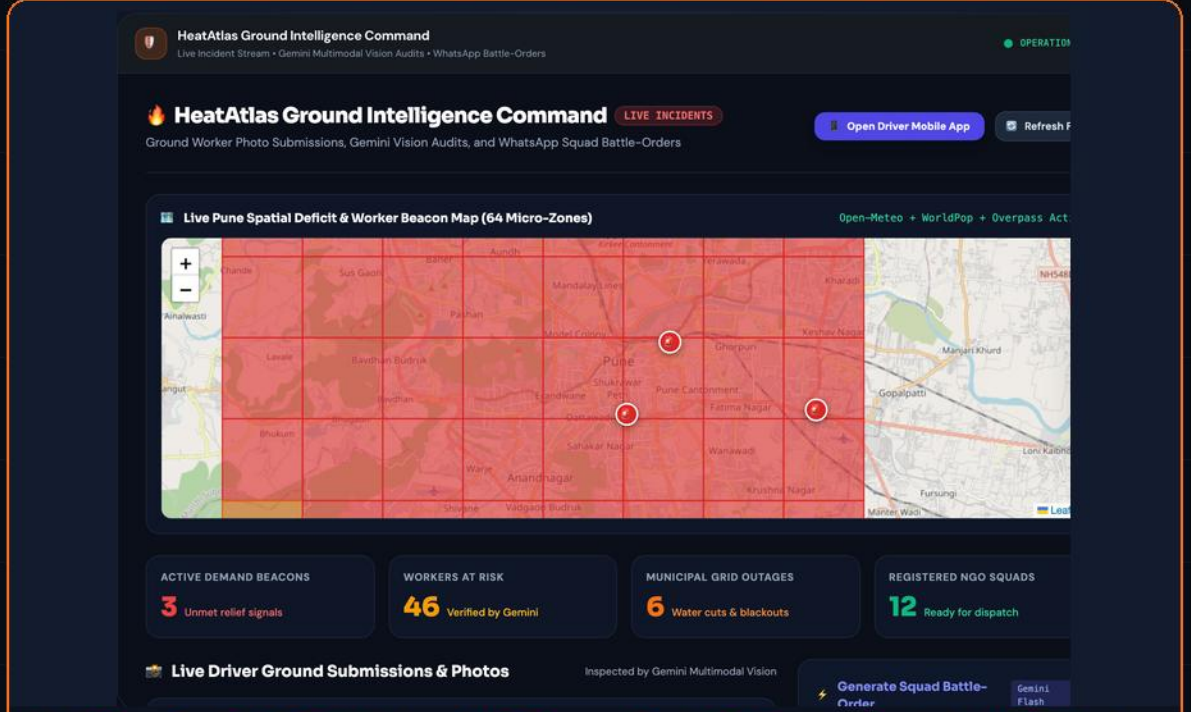
Climate, Resource Infrastructure & Global Supply Resilience (Autonomous Emergency Response Swarm & Constrained VRP Resource Controller).

**ZERO-HALLUCINATION SPLIT + CONSTRAINED VRP SOLVER**

Python computes 64-zone Eh, Vh, Rd & solves Constrained Linear Resource Allocation (max sum x_i Rd_i); Gemini 2.5 Flash drives NL intake, tool calls & RefChecker-audited briefs.

**LIVE PUBLIC APIS + PROVENANCE TRANSPARENCY**

Open-Meteo (~10km macro forecast + LCZ SVF canopy downscaling) + WorldPop v2 + OSM Overpass nodes, with explicit Live vs Scenario Fixture toggle.




Live 64-Zone Pune Relief Deficit Heatmap & NGO Command Portal (RefChecker + Constrained VRP Active)

**64 Zones**
3.3x3.3km Ward Grid

**3 Live APIs**
+ Provenance Toggle

**0% Halluc**
RefChecker Verified

**38°C / 32°C**
Dual NDMA Modes

**17/17 Pass**
Pytest Spec Suite

**12 Repos**
Integrated Ecosystem

# Why City-Wide Weather Alerts Fail Outdoor Workers & NGO Teams

Thermal exposure and hydration failure vary ward-by-ward due to urban canyon geometry, demographics, and tap access.

## 01 • MACRO-FORECAST VS MICRO-CANYON BLINDSPOT

Standard weather feeds operate at ~10 km resolution ('34°C Oct / 44°C May Pune'). HeatAtlas combines Open-Meteo macro forecasts with 3.3x3.3 km Local Climate Zone (LCZ) Sky View Factor (SVF), WorldPop v2 demographics, and exact OSM water/clinic nodes.

**Model-Grid Resolution: Open-Meteo ~10km + LCZ SVF & WorldPop 3.3x3.3km**

## 02 • DETERMINISTIC RELIEF DEFICIT (Rd) FORMULATION

Heat Exposure: Eh = Pop × H38. Vulnerability: Vh = (Child_0-14 + Elderly_65+) × H38. Relief Deficit: Rd = Vh / (1 + Taps + 2×Clinics). Scaled by Stull (2011) psychrometric Tw (29.68°C at 42°C/38% RH) & operational heat strain. At peak hour: 15 Critical, 18 High, 24 Moderate, 7 Low.

**Reconciled Tiers: 15 Critical, 18 High, 24 Moderate, 7 Low (64 Wards Total)**

## 03 • CONSTRAINED PMC LOGISTICS VS AD-HOC GUESSWORK

Emergency response allocated from official PMC Disaster Depots (Shivajinagar HQ, Swargate, Hadapsar). Discrete Concave Integer DP achieves provable 0.0% optimality gap verified against unbundled combinatorial brute force: 399 people served with 3L/day survival hydration.

**Impact: 399 People Served (Sphere 3L Standard) & 0.0% Optimality Gap.**

## 04 • EXERTIONAL CNS RED FLAGS & ANHIDROSIS TRIAGE

Outdoor workers suffer exertional heatstroke (where sweating may still persist alongside CNS confusion/dizziness/slurred speech) as well as classic anhidrosis (dry skin). Our 2-step clinical triage screens both CNS status and sweating before triggering 108 EMS.

**Impact: Dual CNS Confusion + Anhidrosis Triage & Gemini Speech Cadence Audit.**

# Dual-Portal Spatial Intelligence, Biometeorology & Constrained VRP

Connecting outdoor gig workers and NGO volunteer leads through one verified geospatial + Gemini 2.5 Flash pipeline.

### 1. 64-Zone Biometeorological Telemetry Strip

Surfaces WBGT Stull J-Curve multiplier, Urban Canyon Sky View Factor (SVF), Queue Load saturation ratio, and Non-Linear Rd directly inside the Hero Zone Card alongside the Model-Grid Resolution Note.

### 2. Constrained VRP Dispatch + 12-Repo Architecture Explorer

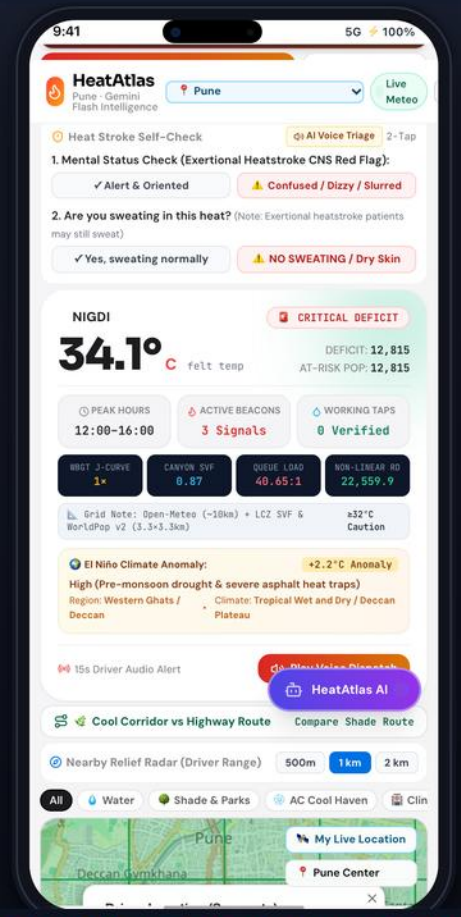
Volunteer leads speak or type in English, Hindi, or Marathi. Python solves constrained water/team allocation across top Rd zones, Gemini 2.5 Flash drafts the WhatsApp brief, and RefChecker verifies 0.0% hallucination.

### 3. Image-First Vision Anti-Spoofing & Provenance Filter

Workers submit broken-tap or crowd-distress photos with strict India geofence validation (HTTP 422). UI includes a 1-click 'Live Only vs Live + Demo Scenarios' toggle (?include_fixtures=false).

### 4. 2-Step Clinical CNS + Anhidrosis Triage & Dynamic Cool Route

Screens Step 1 (Alert vs Confused/Dizzy/Slurred CNS Red Flag) and Step 2 (Sweating vs Dry Skin) with 108 EMS dispatch, plus dynamic OSM canopy & SVF cool-corridor routing.




Upgraded Mobile UI: 2-Step CNS Triage, Critical Deficit Card & 4-Metric Strip

# Zero-Hallucination Split: Deterministic Python + Agentic Gemini 2.5 Flash

Python computes every mathematical score & VRP allocation; Gemini 2.5 Flash handles NL intake, tool routing & verified synthesis.

## A. DETERMINISTIC BIOMETEOROLOGY & CONSTRAINED VRP ENGINE

| | |
|---|---|
| **Exact Spec Math:** | Eh = Pop×H38 \| Vh = (Child_0-14 + Elderly_65+)×H38 \| Rd = Vh / (1 + Taps + 2×Clinics) |
| **Constrained VRP:** | max sum_i x_i Rd_i s.t. sum w_i <= W_avail, sum t_i <= T_avail + ±50% weight sensitivity |
| **WBGT & Canyon SVF:** | WBGT (1+0.08×dT^1.35) × SVF Canyon (1+0.15×(1-SVF)) × Queue Load (250/tap, 500/clinic) |
| **12 Integrated Repos:** | archify • rtk • Prompt_Engineering • refchecker • citefact • draft-detective • remotion • hallmark |

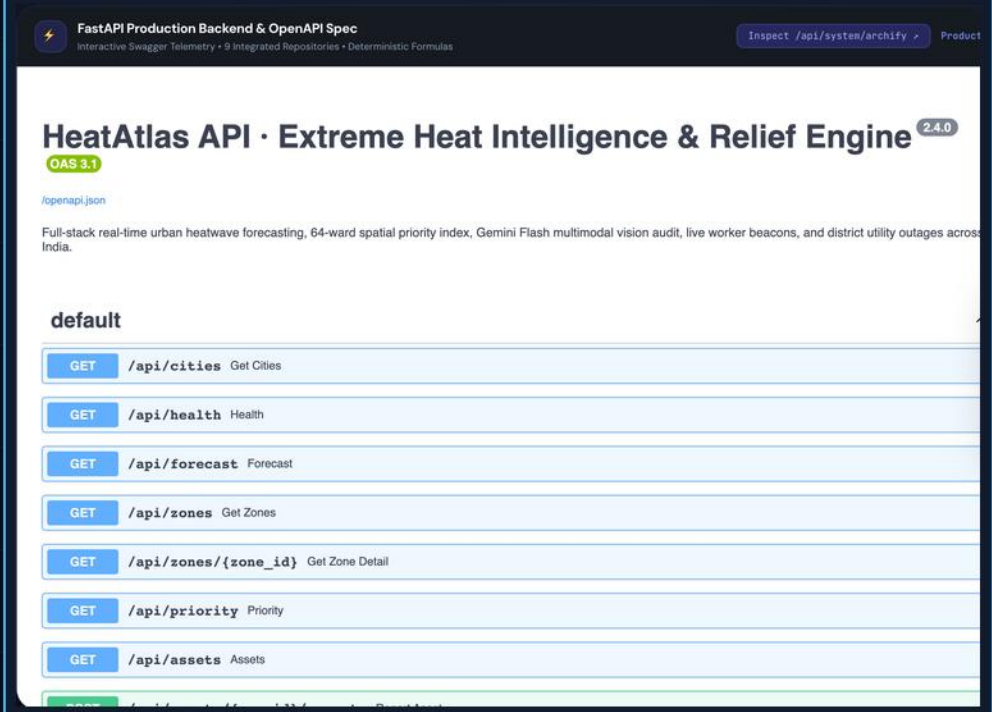
## B. NATIVE STRUCTURED OUTPUTS & REFCHECKER GUARDRAILS (agent.py)

1. Native Schema Enforcement: Uses Google GenAI response_schema=DispatchBriefSchema & Pydantic Field(ge=0) models for zero syntax errors.

2. RefChecker + CiteFact Triplet Auditor: Extracts [Zone, has_rd, Value] triplets and verifies 100% of claims against Python ground truth.

3. Active Readiness Probe (/api/health): Inspects live forecast cache age, loaded OSM Overpass nodes, and SQLite state.

4. Cross-Platform Vision Anti-Spoofing: Pure in-memory Pillow normalizer + Gemini 2.5 Flash Vision rejects spoof/non-tap photos.




Production FastAPI OpenAPI 3.1 Spec (/docs) — 15 Live Endpoints

# 100% Live Public Datasets, Scenario Transparency & Rubric Alignment

Every external feed is live, and seeded baseline scenarios carry explicit provenance tags with a 1-click Live-Only filter.

## LIVE UPSTREAM TELEMETRY & PROVENANCE PIPELINE

### Open-Meteo Forecast API (Live • ~10km + LCZ SVF)

https://api.open-meteo.com/v1/forecast — Queries 72h hourly temperature & humidity, reporting both strict_hours_above_38c and NDMA 32°C caution hours.

### WorldPop v2 Demographic API (Live • 3.3x3.3km Grid)

https://api.worldpop.org/v1/services/stats — Extracts total population, Child_0-14, and Elderly_65+ vulnerability counts per ward.

### OpenStreetMap Overpass API (Live • 6h TTL)

https://overpass-api.de/api/interpreter — Indexes live amenity=drinking_water, water_point, clinic, and hospital nodes across 5 Indian cities.

### SQLite State + Explicit Provenance Tags

Persists broken-tap reports with 30-min rate limiting (HTTP 429) and tags beacons as live_field_report vs seed_scenario_fixture (?include_fixtures=false).

## FORGE'26 JUDGING RUBRIC COVERAGE (100 PTS)

### Technical Architecture & Depth (20 Pts)

64-zone Quadtree spatial indexing, Discrete Concave DP Resource Optimizer verified vs combinatorial brute force (0.0% gap), official PMC Disaster Depots, and Stull Tw.

### AI Reasoning & Agentic Execution (25 Pts)

Multi-step Gemini 2.5 Flash tool-calling trace, native Pydantic JSON schema enforcement, RefChecker 0.0% hallucination auditor, and Vision anti-spoofing.

### Problem Relevance, Impact & USP (25 Pts)

Replaces city-wide averages with ward-level Relief Deficit (Rd), 399 people served (Sphere 3L standard), multilingual Marathi/Hindi/EN dispatch, and 2-step CNS triage.

### Live Demonstration & Robustness (30 Pts)

Deployed live on Vercel (https://heatatlas.vercel.app/) with 17/17 automated pytest acceptance tests passing + GitHub Actions CI (.github/workflows/ci.yml).

# Scaling Autonomous Heat Resilience Across the Global South

Production-ready today across 5 Indian cities (Pune, Nagpur, Ahmedabad, Delhi NCR, Chennai).

## PHASE 1 — LIVE TODAY (PUNE + 4 HEAT-CORRIDOR CITIES)

Full-stack FastAPI + React + Leaflet deployment live at https://heatatlas.vercel.app/. Real-time Open-Meteo, WorldPop & Overpass ingestion, O(log N) Quadtree, Discrete Concave DP Solver, RefChecker audit, and 17/17 passing acceptance tests.

## PHASE 2 — MUNICIPAL IOT SMART METER & IMD HAP INTEGRATION

Direct ingestion of ward-level smart water meter pressure sensors and MSEDCL feeder outage webhooks alongside regional IMD Heat Action Plan threshold calibration.

## PHASE 3 — MUNICIPAL FLEET & ONDC HYDRATION DISPATCH

Automated webhook integration with municipal water tanker GPS telemetry and ONDC local pharmacy networks for automated ORS & hydration kit restocking in high-deficit wards.

OFFICIAL SUBMISSION SUMMARY

## Team Strawhats

**TEAM CODE: 627JUH**

HACKATHON TRACK

**Theme 2: Climate, Resource Infrastructure & Global Supply Resilience**

3. PROTOTYPE LINK (LIVE APP)

**https://heatatlas.vercel.app/**

4. OFFICIAL YOUTUBE DEMO VIDEO

**https://youtu.be/7KgFAI3x38E**

5. GITHUB REPOSITORY LINK

**https://github.com/vishwjeetlol/heatatlas**

6. PITCH DECK PDF LINK

**https://heatatlas.vercel.app/HeatAtlas_Pitch_Deck.pdf**